

Module 1: AI and Risk - Introduction and Overview

100 Questions: Classical AI, Neural Networks, Machine Learning Types, AI Risks

GARP RAI Exam Preparation

QUESTION: What best describes Classical AI?

OPTIONS:

- ❶ AI based on statistical pattern recognition
- ❷ AI based on explicit, precise processing using symbolic logic and formal rules
- ❸ AI that learns from data without explicit programming
- ❹ AI that mimics biological neural networks

ANSWER: B

DETAILED EXPLANATION:

- Classical AI (GOFAI - Good Old-Fashioned AI) is based on explicit, precise processing
- Intelligence is seen as following rules and manipulating symbols
- Typified by symbolic logic and formal rule-governed systems
- Emphasis on logical deduction and search through possibilities

EXAM TRAP: Classical AI = **explicit, rule-based processing!**

QUESTION: What is the Turing Test?

OPTIONS:

- ❶ A test of computing speed
- ❷ A test where a computer proves intelligence by generating indistinguishable conversation
- ❸ A test of memory capacity
- ❹ A test of logical reasoning

ANSWER: B

DETAILED EXPLANATION:

- Proposed by Alan Turing in 1950
- Originally called the "imitation game"
- Computer proves intelligence if textual conversation is indistinguishable from human
- Ranging over any topic chosen by interrogator
- Remains influential and controversial

EXAM TRAP: Turing Test = **indistinguishable conversation**, not true understanding!

QUESTION: What is the difference between specific and general AI?

OPTIONS:

- ❶ Specific AI is faster; general AI is slower
- ❷ Specific AI solves specific problems; general AI has human-level general intelligence
- ❸ Specific AI is newer; general AI is older
- ❹ Specific AI is for business; general AI is for science

ANSWER: B

DETAILED EXPLANATION:

- **Specific (Narrow) AI:** Designed for specific problems, outperforms humans in narrow domains
- **General AI (AGI):** Human-level general intelligence, can handle any intellectual task
- Most practical AI is specific/narrow
- AGI is currently theoretical
- Source of existential risk concerns

EXAM TRAP: Most AI is **narrow/specific**, not general!

QUESTION: What was significant about the 1956 Dartmouth Conference?

OPTIONS:

- ❶ It was the first AI conference
- ❷ The term "artificial intelligence" was coined
- ❸ It established AI as a research field
- ❹ All of the above

ANSWER: D

DETAILED EXPLANATION:

- Term "artificial intelligence" coined by John McCarthy
- Organized by McCarthy, Minsky, Rochester, and Shannon
- Two-month summer conference at Dartmouth College
- Core conjecture: intelligence can be precisely described so machine can simulate it
- Established AI as a research field

EXAM TRAP: Dartmouth Conference (1956) = birth of AI as a field!

QUESTION: What was the Logic Theorist program designed to do?

OPTIONS:

- ❶ Play chess
- ❷ Generate proofs in propositional logic
- ❸ Translate languages
- ❹ Recognize images

ANSWER: B

DETAILED EXPLANATION:

- Developed by Newell, Simon, and Shaw (1955)
- Designed to generate proofs in propositional logic
- Confirmed most theorems in Principia Mathematica
- Found new and more elegant proof of Theorem 2.85
- Early success of classical AI

EXAM TRAP: Logic Theorist = automated theorem proving!

QUESTION: Why were games attractive testbeds for AI development?

OPTIONS:

- ❶ They are simple, rule-governed, and competitive
- ❷ They require no computing resources
- ❸ They are unpredictable
- ❹ They are not rule-governed

ANSWER: A

DETAILED EXPLANATION:

- **Relatively Simple:** Compared to complexities of normal life
- **Straightforwardly Rule-Governed:** Clear regulations for setup, moves, effects, win/loss/draw
- **Competitive:** Players can be rated by skill
- Natural assessment of researcher progress

EXAM TRAP: Games provided **controlled environments** for AI development!

QUESTION: In the Nim game example, how does reinforcement learning work?

OPTIONS:

- ❶ By looking ahead to future moves
- ❷ By adjusting weights based on game outcomes
- ❸ By memorizing all possible moves
- ❹ By following pre-programmed rules

ANSWER: B

DETAILED EXPLANATION:

- Assign weights to each move in each position
- Choose moves randomly based on weights
- Winner: chosen moves positively reinforced, others negatively
- Loser: chosen moves negatively reinforced, others positively
- After 100 games, discovers winning strategy

EXAM TRAP: Reinforcement learning = **learning from trial and error!**

QUESTION: When is lookahead essential in AI?

OPTIONS:

- ❶ When there are few possible situations
- ❷ When forming multistep plans in complex situations
- ❸ When the problem is simple
- ❹ When no planning is needed

ANSWER: B

DETAILED EXPLANATION:

- Essential when forming multistep plans to achieve goals
- Context: Very large number of possible situations overall
- But relatively constrained and predictable options in particular situations
- Examples: Sudoku, Tower of Hanoi, chess

EXAM TRAP: Lookahead = exploring future consequences of actions!

QUESTION: What is the A* algorithm?

OPTIONS:

- ❶ A type of neural network
- ❷ A heuristic search algorithm that finds optimal routes
- ❸ A reinforcement learning algorithm
- ❹ A clustering algorithm

ANSWER: B

DETAILED EXPLANATION:

- Best-known heuristic search technique
- Tracks distance traveled so far
- Estimates distance still to go (optimistic)
- Orders locations by sum of these values
- Explores from location with shortest sum
- Guarantees finding best route if estimate is optimistic

EXAM TRAP: A* = optimal heuristic search!

QUESTION: What is combinatorial explosion?

OPTIONS:

- ❶ A type of neural network architecture
- ❷ Exponential growth in search space as problem size increases
- ❸ A method for speeding up computation
- ❹ A type of machine learning

ANSWER: B

DETAILED EXPLANATION:

- Search space grows enormously as problem size increases
- Relevant numbers of possible choices get multiplied together
- Certainty of solution quickly becomes unfeasible
- Examples: Travelling Salesman, chess, Go
- Earth would be swallowed by Sun before first move

EXAM TRAP: Combinatorial explosion = exponential growth in possibilities!

QUESTION: What is recursion in AI?

OPTIONS:

- ❶ A type of neural network
- ❷ Solving complex problems by reducing to simpler instances of same problem
- ❸ A method for data storage
- ❹ A type of search algorithm

ANSWER: B

DETAILED EXPLANATION:

- Powerful technique in classical AI
- Complex problem solved by reducing to simpler instances
- Tower of Hanoi example: moving n disks reduces to moving $n-1$ disks
- Each subtask reduces to simpler instance
- Eventually reduces to single disk movements

EXAM TRAP: Recursion = solving problems by reducing to simpler versions!

QUESTION: How many steps are required to move n disks in the Tower of Hanoi?

OPTIONS:

- 1 n steps
- 2 n^2 steps
- 3 $2^n - 1$ steps
- 4 $n!$ steps

ANSWER: C

DETAILED EXPLANATION:

- Moving n disks takes $2^n - 1$ steps.
- Moving 9 disks takes 511 steps.
- The result can be proved by mathematical induction.
- Base case: 1 disk = 1 move = $2^1 - 1$.
- Recursive step: Move $n - 1$ disks ($2^{n-1} - 1$), move disk n (1 move), and move $n - 1$ disks ($2^{n-1} - 1$).
- Total: $2(2^{n-1} - 1) + 1 = 2^n - 1$.

EXAM TRAP: Tower of Hanoi takes $2^n - 1$ steps!

QUESTION: What is the minimax principle in adversarial search?

OPTIONS:

- ❶ Minimize your own moves, maximize opponent's moves
- ❷ Presume opponent chooses best reply; choose move minimizing their maximum benefit
- ❸ Always choose the first available move
- ❹ Choose moves randomly

ANSWER: B

DETAILED EXPLANATION:

- In two-player games
- Presume opponent will choose reply maximizing their benefit
- Choose move minimizing that maximum benefit
- Applied recursively at every level
- Also called adversarial search

EXAM TRAP: Minimax = adversarial search principle!

QUESTION: What are heuristics in game playing AI?

OPTIONS:

- ❶ Exact mathematical solutions
- ❷ Calculable criteria or rules of thumb to guide position assessment
- ❸ Random guesses
- ❹ Pre-programmed moves

ANSWER: B

DETAILED EXPLANATION:

- Calculable criteria or rules of thumb
- Guide choice of position to aim for
- Cannot guarantee optimal solution
- Checkers criteria: material balance, mobility, advancement, kings
- Samuel devised 38 criteria

EXAM TRAP: Heuristics = **rules of thumb**, not exact solutions!

QUESTION: How are heuristic weights refined in game-playing AI?

OPTIONS:

- ❶ By human experts
- ❷ By reinforcement learning through playing games
- ❸ By random selection
- ❹ By mathematical optimization

ANSWER: B

DETAILED EXPLANATION:

- Initial weights equal (e.g., 1,000)
- Random variation creates competing strategy
- Play games between current and competing
- Replace current with competing if better
- Repeat iteratively
- Program can learn to beat designer

EXAM TRAP: Weights refined by [reinforcement learning](#)!

QUESTION: What is a major limitation of classical AI?

OPTIONS:

- ❶ It is too fast
- ❷ It is brittle and fails unexpectedly beyond narrow boundaries
- ❸ It requires too much data
- ❹ It cannot handle logic

ANSWER: B

DETAILED EXPLANATION:

- AI systems notoriously brittle
- Fail in unexpected ways beyond familiar boundaries
- Situations not explicitly foreseen
- Representation problem: hard to codify knowledge
- Frame problem: tracking what changes and what doesn't
- Uncertainty: complex probability calculations

EXAM TRAP: Classical AI is brittle - fails beyond narrow boundaries!

QUESTION: What is the frame problem in classical AI?

OPTIONS:

- ❶ A type of neural network
- ❷ Keeping track of what changes and what stays the same when actions are performed
- ❸ A method for image recognition
- ❹ A type of search algorithm

ANSWER: B

DETAILED EXPLANATION:

- Keeping track of what changes when action performed
- Also tracking what stays the same
- Adding to combinatorial explosion
- Makes general search impractical
- Problem for classical AI

EXAM TRAP: Frame problem = tracking what changes and what doesn't!

QUESTION: What are expert systems?

OPTIONS:

- ❶ Neural networks with many layers
- ❷ Systems designed to capture knowledge within specific domains
- ❸ Reinforcement learning algorithms
- ❹ Unsupervised learning methods

ANSWER: B

DETAILED EXPLANATION:

- Designed to capture knowledge in specific domains
- Often use logic programming (PROLOG)
- Response to limitations of general AI
- Knowledge base + inference engine
- But not easily generalized beyond domains

EXAM TRAP: Expert systems = domain-specific knowledge capture!

QUESTION: What is PROLOG?

OPTIONS:

- ❶ A type of neural network
- ❷ A logic programming language used in expert systems
- ❸ A search algorithm
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- Logic programming language
- Used as inferential medium in expert systems
- Part of Japanese Fifth Generation project
- Knowledge base + inference engine architecture
- But limitations in generalization

EXAM TRAP: PROLOG = logic programming for expert systems!

QUESTION: What did Charles Babbage contribute to computing?

OPTIONS:

- ❶ The Turing Test
- ❷ The difference engine and analytical engine concepts
- ❸ The first neural network
- ❹ The first expert system

ANSWER: B

DETAILED EXPLANATION:

- Early 19th century
- Difference engine: mechanical, gear-driven
- Replace error-prone human calculation
- Analytical engine: more ambitious design
- Widely seen as anticipating digital computer
- Never completed

EXAM TRAP: Babbage = early vision of computing!

QUESTION: What is a Turing machine?

OPTIONS:

- ❶ A practical computer built by Turing
- ❷ A theoretical universal programmable computer
- ❸ A type of neural network
- ❹ A search algorithm

ANSWER: B

DETAILED EXPLANATION:

- Theoretical invention (1936)
- Designed to prove fundamental results about mathematics
- Universal programmable computer
- Can follow any given recipe for calculation
- Foundation of computability theory
- Not a practical device

EXAM TRAP: Turing machine = **theoretical**, not practical!

QUESTION: What was Turing's work at Bletchley Park?

OPTIONS:

- ❶ Developing neural networks
- ❷ Decrypting Nazi Enigma codes during WWII
- ❸ Building the first computer
- ❹ Writing AI papers

ANSWER: B

DETAILED EXPLANATION:

- WWII code-breaking
- Mechanical methods of decrypting Enigma codes
- Massive impetus to electronic computer development
- Both sides of Atlantic
- Practical application of computing theory

EXAM TRAP: Bletchley Park = WWII code-breaking!

QUESTION: What was significant about Deep Blue?

OPTIONS:

- ❶ It was the first neural network
- ❷ It defeated Garry Kasparov at chess in 1997
- ❸ It was the first expert system
- ❹ It was the first self-driving car

ANSWER: B

DETAILED EXPLANATION:

- IBM's chess-playing computer
- Defeated Garry Kasparov in 1997
- Kasparov considered strongest human player
- Conspicuous success of classical AI
- Specifically programmed with expert knowledge

EXAM TRAP: Deep Blue = **classical AI triumph!**

QUESTION: What was significant about Samuel's checkers program?

OPTIONS:

- ❶ It was the first neural network
- ❷ It was the first substantial example of machine learning
- ❸ It was the first expert system
- ❹ It was the first search algorithm

ANSWER: B

DETAILED EXPLANATION:

- Developed by Arthur Samuel (late 1950s)
- First substantial example of machine learning
- Samuel coined the term "machine learning"
- Used heuristics and reinforcement learning
- Program learned to play better than designer

EXAM TRAP: Samuel's checkers = **first substantial machine learning!**

QUESTION: In Samuel's checkers program, what are examples of heuristics?

OPTIONS:

- ❶ Material balance, mobility, advancement, kings
- ❷ Random moves
- ❸ All possible moves
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- Material balance (my pieces vs. opponent's)
- Relative mobility of pieces
- How far advanced pieces are
- Relative number of "kings"
- Samuel devised 38 criteria total
- 16 used at any time

EXAM TRAP: Checkers heuristics = material, mobility, advancement, kings!

QUESTION: What did McCulloch and Pitts contribute to neural networks?

OPTIONS:

- ❶ The Perceptron
- ❷ Analysis of neurons in terms of propositional logic
- ❸ Backpropagation
- ❹ Deep learning

ANSWER: B

DETAILED EXPLANATION:

- 1943 paper: "A logical calculus of the ideas immanent in nervous activity"
- Described activity of neurons in brain
- Proposed networks analyzed in terms of propositional logic
- Neural nets equivalent to Turing machines
- Foundation for neural network research

EXAM TRAP: McCulloch-Pitts = **neurons as logic!**

QUESTION: What did Donald Hebb contribute to neural networks?

OPTIONS:

- ❶ The Perceptron
- ❷ Theory of associative learning in neural networks
- ❸ Backpropagation
- ❹ Deep learning

ANSWER: B

DETAILED EXPLANATION:

- 1949 book: "The Organisation of Behaviour"
- Inspired by biology rather than mathematics
- Suggested associative learning in neural networks
- "Neurophysiological postulate" about connection strength
- When cell A repeatedly fires cell B, efficiency increases

EXAM TRAP: Hebb = associative learning postulate!

QUESTION: What was the Perceptron?

OPTIONS:

- ❶ A type of search algorithm
- ❷ A computational model for how brain learns and stores information
- ❸ A type of expert system
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- Developed by Frank Rosenblatt (1958)
- Computational model for learning and memory
- Based on Hebbian learning
- Rejected symbolic/algorithmic approach
- Formulated in terms of probability theory
- Inspired widespread interest in neural networks

EXAM TRAP: Perceptron = [Rosenblatt's neural model!](#)

QUESTION: How does an artificial neuron work?

OPTIONS:

- ❶ Takes inputs, multiplies by weights, sums, applies activation function
- ❷ Randomly generates outputs
- ❸ Stores all inputs in memory
- ❹ Compares inputs to database

ANSWER: A

DETAILED EXPLANATION:

- Takes numerical inputs (i_1, i_2, i_3 , etc.)
- Each input multiplied by weight (w_1, w_2, w_3 , etc.)
- Weights added together (plus bias term w_0)
- Net input fed into activation function f
- Output value v determined
- $v = f(w_0 + w_1 i_1 + w_2 i_2 + w_3 i_3 + \dots)$

EXAM TRAP: Artificial neuron = **weighted sum** + **activation**!

QUESTION: What is the purpose of the activation function in an artificial neuron?

OPTIONS:

- ❶ To store data
- ❷ To introduce non-linearity
- ❸ To reduce computation
- ❹ To increase memory

ANSWER: B

DETAILED EXPLANATION:

- Chosen to be non-linear
- Often approximates step function
- Triggered when net input $\geq$ threshold
- Non-linearity crucial for learning non-linear behavior
- Common: sigmoid, tanh, ReLU

EXAM TRAP: Activation function introduces **non-linearity**!

QUESTION: What is a deep neural network?

OPTIONS:

- ❶ A network with only input and output layers
- ❷ A network with multiple hidden layers between input and output
- ❸ A network with one hidden layer
- ❹ A network with no layers

ANSWER: B

DETAILED EXPLANATION:

- Multiple hidden layers between input and output
- Each neuron connected to all in next layer
- Input layer: represents input data
- Hidden layers: intermediate representations
- Output layer: final prediction
- Learning = changing weights

EXAM TRAP: Deep network = **multiple hidden layers!**

QUESTION: What was the main criticism in Minsky and Papert's "Perceptrons"?

OPTIONS:

- ❶ Perceptrons were too fast
- ❷ Perceptrons couldn't learn XOR and other simple functions
- ❸ Perceptrons required too much data
- ❹ Perceptrons were too complex

ANSWER: B

DETAILED EXPLANATION:

- 1969 book "Perceptrons"
- Fierce critique of single-layer Perceptrons
- Incapable of learning simple logical functions
- Notably "exclusive or" (XOR)
- Later realized: critique applied only to single-layer networks
- Multi-layer networks can learn these functions

EXAM TRAP: Minsky-Papert applied [only to single-layer networks](#)!

QUESTION: What is backpropagation?

OPTIONS:

- ❶ A type of neural network
- ❷ Learning algorithm that works backward through layers to adjust weights
- ❸ A search algorithm
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- Learning algorithm for multi-layer networks
- Rose to prominence in 1980s
- Calculate how weight changes would improve match
- Work back through layers
- Adjust weights by small amount
- Repeat thousands/millions of times
- Uses chain rule of differentiation

EXAM TRAP: Backpropagation = working backward through layers!

QUESTION: What is connectionism?

OPTIONS:

- ❶ A type of search algorithm
- ❷ Approach emphasizing neural networks and distributed processing
- ❸ A type of expert system
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- Displaced pessimism of Minsky and Papert
- Key work: "Parallel Distributed Processing" (1986)
- Showed how layers of simple neurons achieve learning
- Four key attractions: biologically inspired, natural learning, general, distributed
- Enthusiasm declined in 1990s but returned later

EXAM TRAP: Connectionism = **neural network approach!**

QUESTION: What are attractions of connectionism?

OPTIONS:

- ❶ Biologically inspired, natural learning, general, distributed
- ❷ Fast computation, low memory, simple
- ❸ Rule-based, symbolic, logical
- ❹ Deterministic, predictable, transparent

ANSWER: A

DETAILED EXPLANATION:

- **Biologically Inspired:** Potential to shed light on how we think
- **Natural Learning:** Through association and feedback
- **General:** Applied to huge variety of tasks
- **Distributed:** Robust to noisy data, generalizes well, graceful degradation

EXAM TRAP: Connectionism = **biologically inspired, general, distributed!**

QUESTION: What is LeNet-5?

OPTIONS:

- ❶ A type of expert system
- ❷ A convolutional neural network for digit recognition
- ❸ A search algorithm
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- Developed by Yann LeCun and colleagues (1998)
- Designed for recognition of digits
- 8 layers
- Convolutional neural network
- Input: 32×32 pixel image
- Total: 9,118 neurons

EXAM TRAP: LeNet-5 = convolutional network for digits!

QUESTION: What is the purpose of convolutional layers?

OPTIONS:

- ❶ To reduce computation
- ❷ To enable efficient identification of local features in an image
- ❸ To store data
- ❹ To increase memory

ANSWER: B

DETAILED EXPLANATION:

- Apply local filter (kernel) repeatedly across grid
- Filter = small matrix of weights
- Weights multiplied by pixel activation values
- Sum of products = activation in convolutional layer
- Weights learned like other network weights
- Efficient identification of local features

EXAM TRAP: Convolutional layers = **local feature identification!**

QUESTION: What was significant about AlexNet?

OPTIONS:

- ❶ It was the first neural network
- ❷ It won ImageNet 2012 with 15.3% error rate
- ❸ It was the first expert system
- ❹ It was the first search algorithm

ANSWER: B

DETAILED EXPLANATION:

- Developed by Krizhevsky, Sutskever, and Hinton (2012)
- Won ImageNet Large Scale Visual Recognition Challenge
- Error rate: 15.3% (vs. 25.8% in 2011)
- Nearly 100 times bigger than LeNet-5
- Almost 900,000 neurons
- Sparked renewed interest in deep learning

EXAM TRAP: AlexNet = **deep learning breakthrough!**

QUESTION: How did GPUs enable the deep learning revolution?

OPTIONS:

- ❶ By reducing memory requirements
- ❷ By enabling parallel processing of image pixels
- ❸ By simplifying algorithms
- ❹ By reducing data needs

ANSWER: B

DETAILED EXPLANATION:

- Developed for animated computer games
- Process image pixels in parallel
- Up to 10,000 cores
- Perfect for matrix operations in neural networks
- Made AlexNet possible
- Accelerated training

EXAM TRAP: GPUs = parallel processing for neural networks!

QUESTION: What was remarkable about DeepMind's Atari program?

OPTIONS:

- ❶ It was given all game rules
- ❷ It learned to play from scratch with no game information
- ❸ It used expert knowledge
- ❹ It was programmed move by move

ANSWER: B

DETAILED EXPLANATION:

- No information about goal of games
- No information about significance of screen images
- No information about effect of user actions
- Learned entirely from trying actions
- Based on pixel information and game score
- Performed better than expert human

EXAM TRAP: Atari = self-taught from scratch!

QUESTION: What was significant about AlphaGo?

OPTIONS:

- ❶ It was the first neural network
- ❷ It defeated World Go Champion Lee Sedol
- ❸ It was the first expert system
- ❹ It was the first search algorithm

ANSWER: B

DETAILED EXPLANATION:

- DeepMind's deep-learning program
- Defeated Fan Hui 5-0 (2014)
- Defeated Lee Sedol 4-1 (2015)
- Go considered too subtle for computers
- Historic milestone

EXAM TRAP: AlphaGo = **mastering** Go!

QUESTION: What is significant about AlphaZero?

OPTIONS:

- ❶ It was programmed by human experts
- ❷ It learned by playing against itself without human input
- ❸ It used game databases
- ❹ It was the first neural network

ANSWER: B

DETAILED EXPLANATION:

- More generalized program
- Capable of teaching itself new games
- Given: starting position, moves, rules
- No human expert input
- No game databases
- Self-trained in hours
- Defeated Stockfish

EXAM TRAP: AlphaZero = self-teaching through self-play!

QUESTION: What is the inscrutability of deep learning?

OPTIONS:

- ❶ The network is too fast
- ❷ Information stored implicitly in weights, impossible to interpret
- ❸ The network requires too much data
- ❹ The network is too simple

ANSWER: B

DETAILED EXPLANATION:

- Information stored implicitly within network weights
- Complex pattern of weights represents learned information
- Impossible for unaided human to interpret
- Behavior predictable only by experience
- "Black box" nature
- No explanation of decisions

EXAM TRAP: Inscrutability = implicit, incomprehensible representations!

QUESTION: How is deep learning similar to human cognition?

OPTIONS:

- ❶ Both use explicit calculation
- ❷ Both use unconscious pattern recognition
- ❸ Both use symbolic logic
- ❹ Both use random guessing

ANSWER: B

DETAILED EXPLANATION:

- Closer to unconscious pattern-recognition
- Unlike explicit calculation or reasoning
- We learn by examples without awareness
- Hard to articulate how we recognize things
- Example: Recognizing dog or letter "f"

EXAM TRAP: Deep learning = unconscious pattern recognition!

QUESTION: Do deep learning machines "understand" what they are doing?

OPTIONS:

- ❶ Yes, they have conscious understanding
- ❷ No, they are completely non-sentient
- ❸ Yes, they have awareness
- ❹ No, but they have feelings

ANSWER: B

DETAILED EXPLANATION:

- Machines do NOT "understand" reflectively
- Completely non-sentient
- No awareness or conscious understanding
- No reflective understanding
- Just pattern recognition

EXAM TRAP: Deep learning = non-sentient pattern recognition!

QUESTION: How does ChatGPT work?

OPTIONS:

- ❶ By understanding the meaning of text
- ❷ By predicting most likely words based on statistical patterns
- ❸ By following logical rules
- ❹ By searching a database

ANSWER: B

DETAILED EXPLANATION:

- Statistical analysis on 300 billion words
- Records probability that word sequences will continue
- Predicts most likely words to follow
- Chooses words (with some randomness)
- Adds word to text, repeats
- Based on imitating training data

EXAM TRAP: ChatGPT = [statistical word prediction](#)!

QUESTION: Why does ChatGPT make errors in chess?

OPTIONS:

- ❶ It doesn't know the rules
- ❷ It has no internal model of the board position
- ❸ It is too slow
- ❹ It requires too much data

ANSWER: B

DETAILED EXPLANATION:

- No internal model of board position
- No mechanisms of analysis or lookahead
- Stores textual patterns, not domain knowledge
- Subject to absurd errors
- Apparent intelligence based on illusion

EXAM TRAP: ChatGPT = no domain model, just text patterns!

QUESTION: What does "stochastic parrot" mean?

OPTIONS:

- ❶ A type of neural network
- ❷ A system that generates plausible text without understanding
- ❸ A search algorithm
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- Term coined by Emily M. Bender (2021)
- Describes LLMs like ChatGPT
- Generates plausible text without understanding
- Based on statistical patterns
- No internal model of domain

EXAM TRAP: Stochastic parrot = [mimicking without understanding!](#)

QUESTION: What is Reinforcement Learning from Human Feedback (RLHF)?

OPTIONS:

- ❶ A type of neural network
- ❷ Using human assessments to generate reward model for rating responses
- ❸ A search algorithm
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- System generates range of responses
- Humans assess relative suitability
- Statistical analysis of feedback
- Generate "reward model"
- Use reward model to rate responses
- Select most appropriate responses

EXAM TRAP: RLHF = using human feedback to improve LLMs!

QUESTION: What is supervised fine-tuning?

OPTIONS:

- ❶ A type of neural network
- ❷ Training on human-crafted responses to typical prompts
- ❸ A search algorithm
- ❹ A clustering method

ANSWER: B

DETAILED EXPLANATION:

- Collect typical prompts
- Human experts craft suitable responses
- Train model on these examples
- Model learns to generate similar responses
- First stage of training
- Followed by RLHF

EXAM TRAP: Fine-tuning = learning from human examples!

QUESTION: What are the four main types of machine learning?

OPTIONS:

- ❶ Supervised, unsupervised, reinforcement, semi-supervised
- ❷ Neural, symbolic, statistical, probabilistic
- ❸ Deep, shallow, wide, narrow
- ❹ Online, offline, batch, streaming

ANSWER: A

DETAILED EXPLANATION:

- **Reinforcement Learning:** Sequence of actions, trial and error
- **Supervised Learning:** Labelled examples, categorize or predict
- **Unsupervised Learning:** Identify patterns without labels
- **Semi-Supervised Learning:** Combination of supervised and unsupervised

EXAM TRAP: Four types = reinforcement, supervised, unsupervised, semi-supervised!

QUESTION: What is reinforcement learning?

OPTIONS:

- ❶ Learning from labelled examples
- ❷ Learning from trial and error with feedback
- ❸ Finding patterns without labels
- ❹ Learning from unlabeled data

ANSWER: B

DETAILED EXPLANATION:

- Sequence of actions or decisions
- Changing environment
- Trial and error
- Positive/negative feedback
- Examples: Nim, Atari, AlphaGo

EXAM TRAP: Reinforcement learning = learning from experience with feedback!

QUESTION: What is supervised learning?

OPTIONS:

- ❶ Learning from labelled examples
- ❷ Learning from trial and error
- ❸ Finding patterns without labels
- ❹ Learning from unlabeled data

ANSWER: A

DETAILED EXPLANATION:

- Learn from labelled examples
- Categorize or predict
- Training data with labels
- Deep learning particularly risky
- Inscrutability and hidden biases

EXAM TRAP: Supervised learning = learning from labelled examples!

QUESTION: What is unsupervised learning?

OPTIONS:

- ❶ Learning from labelled examples
- ❷ Learning from trial and error
- ❸ Finding patterns without labels
- ❹ Learning from labelled data

ANSWER: C

DETAILED EXPLANATION:

- Identify patterns without labels
- No human input in form of labels
- Cluster analysis
- Dimensionality reduction
- Principal Component Analysis (PCA)

EXAM TRAP: Unsupervised learning = finding patterns without labels!

QUESTION: What is semi-supervised learning?

OPTIONS:

- ❶ Learning from fully labelled data
- ❷ Combination of supervised and unsupervised learning with partial labels
- ❸ Learning without any labels
- ❹ Learning from trial and error

ANSWER: B

DETAILED EXPLANATION:

- Only part of data is labelled
- Unlabeled data for patterns
- Pseudo-labels determined by estimating model
- Combination of supervised and unsupervised

EXAM TRAP: Semi-supervised = **partial labels!**

QUESTION: What are applications of reinforcement learning?

OPTIONS:

- ① Robotics, healthcare, finance, optimization
- ② Image classification, speech recognition
- ③ Document clustering, dimensionality reduction
- ④ Text translation, sentiment analysis

ANSWER: A

DETAILED EXPLANATION:

- **Robotics:** Teaching robots/autonomous vehicles
- **Healthcare:** Monitoring treatment effects
- **Finance:** Trading algorithms
- **Optimization:** Any area where optimization by learning from experience

EXAM TRAP: RL applications = **robotics, healthcare, finance, optimization!**

QUESTION: What are applications of supervised learning?

OPTIONS:

- ① Robotics, trading algorithms
- ② Image classification, speech recognition, machine translation
- ③ Document clustering, dimensionality reduction
- ④ All of the above

ANSWER: B

DETAILED EXPLANATION:

- **Image Classification:** Handwriting, faces, road signs
- **Speech Recognition**
- **Machine Translation**
- **Recommendation Systems**
- **Prediction:** Stock prices, house prices, credit scoring

EXAM TRAP: Supervised applications = **classification, recognition, prediction!**

QUESTION: What are applications of unsupervised learning?

OPTIONS:

- ① Robotics, trading algorithms
- ② Image classification, speech recognition
- ③ Cluster analysis, dimensionality reduction
- ④ All of the above

ANSWER: C

DETAILED EXPLANATION:

- **Cluster Analysis:** Groups of similar objects
- **Dimensionality Reduction:** Main dimensions of variation
- **Principal Component Analysis (PCA)**
- **Applications:** Customer segmentation, document clustering

EXAM TRAP: Unsupervised applications = clustering, dimensionality reduction!

QUESTION: What is Principal Component Analysis (PCA)?

OPTIONS:

- ❶ A supervised learning method
- ❷ A technique for dimensionality reduction and cluster analysis
- ❸ A reinforcement learning method
- ❹ A type of neural network

ANSWER: B

DETAILED EXPLANATION:

- Classical method for dimensionality reduction
- Identifies main dimensions of variation
- Enables easier grasp of range and relationships
- Highlights clusters of similar items
- Less inscrutable than deep learning

EXAM TRAP: PCA = finding most informative dimensions!

QUESTION: In the village example, what does PCA reveal?

OPTIONS:

- ❶ The height of houses
- ❷ The most informative presentation of village layout
- ❸ The color of houses
- ❹ The age of houses

ANSWER: B

DETAILED EXPLANATION:

- Original coordinates: latitude, longitude, height
- New framework aligns with best discrimination
- First vector: parallel to road (most spacing)
- View from above: most information
- Most informative presentation of layout

EXAM TRAP: PCA finds **most informative dimensions!**

QUESTION: In Burrows' stylometry example, what did PCA reveal?

OPTIONS:

- ❶ The meaning of the texts
- ❷ Clustering of authors and attribution of "Memoirs" to Smollett
- ❸ The publication dates
- ❹ The length of texts

ANSWER: B

DETAILED EXPLANATION:

- 16 texts in 50-dimensional space
- Coordinates proportional to word frequencies
- Texts close together: similar patterns
- PCA finds best 2D projection
- Female authors clustered together
- "Memoirs" close to Smollett

EXAM TRAP: PCA clusters similar texts without labels!

QUESTION: What is cluster analysis?

OPTIONS:

- ❶ Learning from labelled examples
- ❷ Groups of similar objects identified without training set
- ❸ Learning from trial and error
- ❹ A type of neural network

ANSWER: B

DETAILED EXPLANATION:

- Unsupervised learning technique
- Groups of similar objects identified
- Without training set
- Automatic identification
- Applications: customer segmentation, document clustering

EXAM TRAP: Cluster analysis = automatic grouping of similar objects!

QUESTION: What is dimensionality reduction?

OPTIONS:

- ❶ Learning from labelled examples
- ❷ Identifying main dimensions of variation to simplify data
- ❸ Learning from trial and error
- ❹ A type of neural network

ANSWER: B

DETAILED EXPLANATION:

- Unsupervised learning technique
- Identifies main dimensions of variation
- Enables easier grasp of range and relationships
- Simplify complex data
- Methods: PCA, autoencoders, t-SNE, UMAP

EXAM TRAP: Dimensionality reduction = simplifying complex data!

QUESTION: What are risks of inscrutability?

OPTIONS:

- ❶ Hidden bias, privacy risks, difficulty of eradication
- ❷ Faster computation, lower memory
- ❸ Simpler models, easier interpretation
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **Hidden Bias:** Bias in training data learned and perpetuated
- **Privacy Risks:** Complex information holds clues to personal characteristics
- **Difficulty of Eradication:** Hard to identify source of hidden biases
- Deep network provides no explanation
- True explanation hidden in vast web of weights

EXAM TRAP: Inscrutability = hidden bias, privacy, hard to eradicate!

QUESTION: How does hidden bias arise in deep learning?

OPTIONS:

- ❶ From explicit programming
- ❷ From bias in training data labels learned by model
- ❸ From random errors
- ❹ From hardware failures

ANSWER: B

DETAILED EXPLANATION:

- Bias in labelling of training data
- Historical prejudice
- Learned by model
- Perpetuated in outputs
- Hidden and hard to eradicate
- Proxy variables complicate

EXAM TRAP: Hidden bias = bias in training data!

QUESTION: Why is removing explicit bias insufficient?

OPTIONS:

- ❶ Because bias is always explicit
- ❷ Because proxy variables correlate with protected characteristics
- ❸ Because data is always clean
- ❹ Because models are always fair

ANSWER: B

DETAILED EXPLANATION:

- Lots of features correlate with protected characteristics
- Subjects at school, sports, hobbies, careers
- Postal codes correlate with race
- Removing explicit information insufficient
- Models learn proxy correlations

EXAM TRAP: Proxy variables **perpetuate bias** even without explicit information!

QUESTION: How can AI threaten privacy?

OPTIONS:

- ❶ By storing data insecurely
- ❷ By inferring sensitive personal characteristics from data
- ❸ By sharing data with third parties
- ❹ All of the above

ANSWER: B

DETAILED EXPLANATION:

- Deep implicit entanglement of information
- Network holds clues to personal characteristics
- Facebook Likes predict sexual orientation, ethnicity, etc.
- Risks of manipulation
- Targeted advertising

EXAM TRAP: AI can **infer sensitive characteristics!**

QUESTION: What is the risk of over-reliance on AI?

OPTIONS:

- ❶ Increased autonomy
- ❷ Automation bias, complacency, reduced vigilance
- ❸ Better decision making
- ❹ Improved judgment

ANSWER: B

DETAILED EXPLANATION:

- Automation bias
- Complacency and reduced vigilance
- Undermines autonomy
- Neglect to develop own judgment
- Fail to realize when system is wrong

EXAM TRAP: Over-reliance = automation bias!

QUESTION: What is automation bias?

OPTIONS:

- ❶ Distrust of automated systems
- ❷ Tendency to over-rely on automated systems
- ❸ Preference for manual methods
- ❹ Ignoring technology

ANSWER: B

DETAILED EXPLANATION:

- Tendency of humans to over-rely on automated systems
- Leads to complacency and reduced vigilance
- Errors of oversight
- Especially problematic in critical situations
- Skill atrophy

EXAM TRAP: Automation bias = **over-reliance on automated systems!**

QUESTION: What are adversarial examples?

OPTIONS:

- ❶ Random noise
- ❷ Deliberately designed inputs causing errors in deep learning
- ❸ Natural variations
- ❹ Data entry errors

ANSWER: B

DETAILED EXPLANATION:

- Deliberately designed inputs
- Cause deep learning models to make errors
- Often imperceptible to humans
- Example: sticker on road sign
- Stop sign misidentified as speed limit sign
- Security vulnerability

EXAM TRAP: Adversarial examples = imperceptible changes cause errors!

QUESTION: What is hallucination in LLMs?

OPTIONS:

- ❶ Accurate information
- ❷ False but plausible information
- ❸ Missing information
- ❹ Delayed information

ANSWER: B

DETAILED EXPLANATION:

- Generate textual information completely false
- Yet often plausible
- Inevitable limitation
- Based on statistical patterns
- Not on understanding of truth
- Hard to detect

EXAM TRAP: Hallucination = false but plausible information!

QUESTION: What is existential risk from AI?

OPTIONS:

- ❶ Risk of job losses
- ❷ Fear of superintelligence threatening humanity
- ❸ Risk of privacy breaches
- ❹ Risk of bias

ANSWER: B

DETAILED EXPLANATION:

- Fear of "existential" risk to humanity
- Especially superintelligence
- AI making decisions against interests of creators
- Autonomous weapons systems
- No consensus on nature or extent

EXAM TRAP: Existential risk = **no consensus on nature or extent!**

QUESTION: What are risks to individuals from AI?

OPTIONS:

- ❶ Faulty decisions, automation bias, manipulation, privacy threats
- ❷ Job losses, inequality
- ❸ Reputational damage, regulatory fines
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **Faulty Decisions:** Poorly designed AI systems
- **Automation Bias:** Over-reliance
- **Manipulation:** Algorithms drawing on information
- **Privacy Threats:** Inferences made from information

EXAM TRAP: Individual risks = faulty decisions, automation bias, manipulation, privacy!

QUESTION: What are risks to organizations from AI?

OPTIONS:

- ① Operational, reputational, regulatory, legal
- ② Job losses, inequality
- ③ Faulty decisions, manipulation
- ④ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **Operational Risk:** Over-reliance, bad decision making
- **Reputational Risk:** AI practices hurting individuals
- **Regulatory Risk:** Existing privacy laws apply
- **Legal Risk:** Lawsuits, liability

EXAM TRAP: Organizational risks = operational, reputational, regulatory, legal!

QUESTION: What are risks to society from AI?

OPTIONS:

- ❶ Job losses, inequality, misinformation, existential risk
- ❷ Faulty decisions, manipulation
- ❸ Operational, reputational
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **Economic Displacement:** Job losses, widening wealth gap
- **Global Inequality:** Between countries
- **Misinformation:** Deep fakes, influence public opinion
- **Existential Risk:** Superintelligence concerns

EXAM TRAP: Societal risks = displacement, inequality, misinformation, existential!

QUESTION: Why is training data quality important?

OPTIONS:

- ❶ It doesn't matter
- ❷ Models only as good as training data
- ❸ Quality only affects speed
- ❹ Quality only affects memory

ANSWER: B

DETAILED EXPLANATION:

- Models only as good as training data
- Garbage in, garbage out
- Quality determines performance
- Key attributes: accuracy, completeness, consistency
- Common issues: missing values, outliers, noise, bias

EXAM TRAP: Training data quality **determines model quality!**

QUESTION: What is data leakage?

OPTIONS:

- ❶ Sharing data with third parties
- ❷ Using future or concurrent data that wouldn't be observable at training time
- ❸ Losing data
- ❹ Corrupting data

ANSWER: B

DETAILED EXPLANATION:

- Using future or concurrent data
- Wouldn't have been observable at training time
- Causes optimistic performance estimates
- Model appears accurate but fails in production
- Prevention: careful data splitting, time-based validation

EXAM TRAP: Data leakage = using future data in training!

QUESTION: What is model robustness?

OPTIONS:

- ❶ Model speed
- ❷ How well model performs on unseen data or changes
- ❸ Model size
- ❹ Model complexity

ANSWER: B

DETAILED EXPLANATION:

- Measure of how well model performs
- With inputs previously not encountered
- Or changes in data
- Deep learning vulnerable to adversarial examples
- Improvement: testing, adversarial training, ensembles

EXAM TRAP: Robustness = performing well on unseen data!

QUESTION: What is model interpretability?

OPTIONS:

- ❶ Model speed
- ❷ Degree to which human can comprehend model behavior
- ❸ Model size
- ❹ Model accuracy

ANSWER: B

DETAILED EXPLANATION:

- Degree to which human can comprehend model behavior
- With built-in mechanisms
- Understand how inputs affect outputs
- Spectrum: decision trees high, neural networks low
- Trade-off: accuracy vs. interpretability

EXAM TRAP: Interpretability vs. accuracy **trade-off!**

QUESTION: What is model explainability?

OPTIONS:

- ❶ Model speed
- ❷ Capacity to explain how AI makes decisions after the fact
- ❸ Model size
- ❹ Model accuracy

ANSWER: B

DETAILED EXPLANATION:

- Capacity to explain how AI makes decisions
- Often after the fact (ex-post)
- Understandable terms
- Importance: accountability, trust, error identification
- Challenges: deep networks opaque, billions of weights

EXAM TRAP: Explainability = explaining decisions after the fact!

QUESTION: What is feature engineering?

OPTIONS:

- ❶ Building hardware
- ❷ Selecting and engineering appropriate features for model
- ❸ Writing code
- ❹ Testing models

ANSWER: B

DETAILED EXPLANATION:

- Selecting and engineering appropriate features
- Variables used as inputs for model
- Critical for model performance
- Activities: transforming, combining, creating derived features
- Examples: log transformations, polynomial features

EXAM TRAP: Feature engineering = **creating informative inputs!**

QUESTION: What is model selection?

OPTIONS:

- ❶ Choosing hardware
- ❷ Choosing modeling approach after evaluating alternatives
- ❸ Writing code
- ❹ Testing models

ANSWER: B

DETAILED EXPLANATION:

- Choosing modeling approach
- Comprehensive evaluation of alternatives
- Documenting rationale
- Considerations: problem type, data, interpretability, resources
- Approaches: start simple, compare, cross-validate

EXAM TRAP: Model selection requires **systematic evaluation**!

QUESTION: What are hyperparameters?

OPTIONS:

- ❶ Parameters learned from data
- ❷ Configuration settings chosen before training
- ❸ Model outputs
- ❹ Data features

ANSWER: B

DETAILED EXPLANATION:

- Parameters not learned from data
- Set before training
- Configuration settings
- Examples: learning rate, number of layers, regularization
- Methods: grid search, random search, Bayesian optimization

EXAM TRAP: Hyperparameters = settings chosen before training!

QUESTION: What is k-fold cross-validation?

OPTIONS:

- ❶ Training on all data
- ❷ Splitting data into k folds, training on k-1, validating on 1
- ❸ Testing on training data
- ❹ Using no validation

ANSWER: B

DETAILED EXPLANATION:

- Split data into k folds
- Train on k-1 folds
- Validate on 1 fold
- Repeat k times
- Average results
- Benefits: efficient use, reliable estimate, reduces overfitting

EXAM TRAP: Cross-validation = **train on k-1, validate on 1!**

QUESTION: What are types of model testing?

OPTIONS:

- ❶ Unit, component, integration, system, performance, regression, acceptance, adversarial
- ❷ Only unit tests
- ❸ Only system tests
- ❹ Only acceptance tests

ANSWER: A

DETAILED EXPLANATION:

- **Unit:** Individual code pieces
- **Component:** Individual code sections
- **Integration:** Interfaces between components
- **System:** Complete integrated system
- **Performance:** Speed and stability
- **Regression:** After modifications
- **Acceptance:** Ready for deployment
- **Adversarial:** Extreme scenarios

EXAM TRAP: Testing should be **comprehensive** and **start early**!

QUESTION: What is white-box testing?

OPTIONS:

- ❶ Testing without code access
- ❷ Testing with access to internal code and structures
- ❸ Testing from user perspective
- ❹ Testing only outputs

ANSWER: B

DETAILED EXPLANATION:

- Testers have access to internal data structures
- Access to algorithms
- Access to actual code
- Testing at unit level
- From developer's perspective
- Line-by-line proofreading
- More effective for finding specific issues

EXAM TRAP: White-box = [access to internal code](#)!

QUESTION: What is black-box testing?

OPTIONS:

- ❶ Testing with code access
- ❷ Treating software as closed box, from user perspective
- ❸ Testing internal structures
- ❹ Line-by-line proofreading

ANSWER: B

DETAILED EXPLANATION:

- Treats software as closed box
- From user's perspective
- No knowledge of internal implementation
- Reduces likelihood of bias
- Functional testing
- Suitable for proprietary models

EXAM TRAP: Black-box = [user perspective](#), [no code access](#)!

QUESTION: What is gray-box testing?

OPTIONS:

- ❶ Testing with full code access
- ❷ Testing with knowledge of architecture but not code
- ❸ Testing without any knowledge
- ❹ Testing only outputs

ANSWER: B

DETAILED EXPLANATION:

- Between white-box and black-box
- Informed by knowledge of functional specifications
- Knowledge of architecture
- Does not access code itself
- Combines benefits of both approaches
- Suitable for distributed applications

EXAM TRAP: Gray-box = **knowledge of architecture, not code!**

QUESTION: What is the use test?

OPTIONS:

- ❶ Testing model speed
- ❷ Examining model within context, considering human interaction and usage
- ❸ Testing model accuracy
- ❹ Testing model size

ANSWER: B

DETAILED EXPLANATION:

- Examines model within context
- Considers human interaction
- Assesses actual usage
- Evaluates acceptance
- Focuses on people, not numbers
- Presented to senior management
- Qualitative in nature

EXAM TRAP: Use test focuses on [how people use the model](#)!

QUESTION: Why should there be limits on model use?

OPTIONS:

- ❶ To make models harder to use
- ❷ To prevent misuse beyond intended application
- ❸ To reduce model count
- ❹ To increase model risk

ANSWER: B

DETAILED EXPLANATION:

- Every model based on assumptions
- Assumptions should be clearly stated
- Identify limitations and weaknesses
- Prevent misuse
- Types: product applicability, access restrictions, portfolio size

EXAM TRAP: Model limits prevent misuse beyond intended application!

QUESTION: What factors determine model risk ranking?

OPTIONS:

- ❶ Materiality, complexity, exposure
- ❷ Speed, cost, accuracy
- ❸ Size, weight, color
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **Materiality:** Economic, operational, reputational consequences
- **Complexity:** May elevate ranking even if materiality low
- **Exposure:** Published financial reports, wide impact
- Minimum standards for all models
- Regardless of materiality

EXAM TRAP: Risk ranking considers **materiality, complexity, exposure!**

QUESTION: What are the three lines of defense in model risk management?

OPTIONS:

- ❶ Business/owners, risk management, internal audit
- ❷ Developers, testers, users
- ❸ Management, employees, customers
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **First Line:** Business and model owners/developers
- **Second Line:** Independent risk management/validation
- **Third Line:** Internal audit
- Provide checks and balances
- Clear communication protocols

EXAM TRAP: Three lines = business, risk management, audit!

QUESTION: What is model validation?

OPTIONS:

- ❶ Building models
- ❷ Rigorously testing model, inputs, outputs, governance to identify residual risk
- ❸ Using models
- ❹ Documenting models

ANSWER: B

DETAILED EXPLANATION:

- Rigorously testing model, inputs, outputs, governance
- Identify extent of residual model risk
- Assess mitigating controls
- Initial, periodic, change-based
- No "valid" model - attempt to invalidate

EXAM TRAP: Validation = attempt to invalidate, not prove validity!

QUESTION: What is the model change management process?

OPTIONS:

- ❶ Change requests, impact assessment, documentation, validation, approval
- ❷ Only documentation
- ❸ Only approval
- ❹ Only implementation

ANSWER: A

DETAILED EXPLANATION:

- Change requests
- Impact assessment
- Documentation
- Validation
- Approval
- Implementation
- Monitoring

EXAM TRAP: Model changes require **structured governance**!

QUESTION: What is data governance?

OPTIONS:

- ❶ Collecting as much data as possible
- ❷ System of decision rights and accountabilities for information-related processes
- ❸ Storing data indefinitely
- ❹ Sharing data widely

ANSWER: B

DETAILED EXPLANATION:

- System of decision rights and accountabilities
- For information-related processes
- People, processes, technologies
- Key components: strategy, quality, provenance, classification, security
- Three principles: consent, minimization, data subject rights

EXAM TRAP: Data governance = decision rights and accountabilities!

QUESTION: Why is GenAI governance harder?

OPTIONS:

- ❶ Opacity, unpredictability, emergent capabilities, post-deployment shifts
- ❷ It is simpler than traditional AI
- ❸ It requires no governance
- ❹ It is always transparent

ANSWER: A

DETAILED EXPLANATION:

- Opacity and unpredictability
- Emergent capabilities
- Post-deployment shifts
- Embedded in workflows
- Fragmented regulation
- Requires internal, external, adaptive approaches

EXAM TRAP: GenAI governance requires [internal](#), [external](#), [adaptive approaches](#)!

QUESTION: What is responsible AI?

OPTIONS:

- ❶ Marketing term
- ❷ Ethical and responsible development, deployment, use of AI systems
- ❸ A type of algorithm
- ❹ A regulatory framework

ANSWER: B

DETAILED EXPLANATION:

- Ethical and responsible development, deployment, use
- Benefit society holistically
- Mitigate potential risks
- Align with ethical standards, industry standards, regulation
- Fair, transparent, accountable operation

EXAM TRAP: Responsible AI = ethical development and deployment!

QUESTION: What are five common AI ethics principles?

OPTIONS:

- ❶ Nonmaleficence, beneficence, justice, autonomy, explainability
- ❷ Speed, cost, accuracy, efficiency, profit
- ❸ Privacy, security, transparency, accountability, fairness
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **Nonmaleficence:** Do no harm
- **Beneficence:** Do good
- **Justice:** Fairness
- **Autonomy:** Respect for persons
- **Explainability:** Transparency

EXAM TRAP: Five principles = nonmaleficence, beneficence, justice, autonomy, explainability!

QUESTION: What are key AI regulations?

OPTIONS:

- ❶ GDPR, DMA, DSA, EU AI Act
- ❷ Only GDPR
- ❸ Only EU AI Act
- ❹ None of the above

ANSWER: A

DETAILED EXPLANATION:

- **GDPR:** General Data Protection Regulation
- **DMA:** Digital Markets Act
- **DSA:** Digital Services Act
- **EU AI Act:** Comprehensive AI regulation
- Risk-based approach

EXAM TRAP: EU has **comprehensive AI regulation!**

QUESTION: What is the most comprehensive summary of AI and risk?

OPTIONS:

- ❶ AI is risk-free
- ❷ AI has benefits but also significant risks requiring governance, ethics, and regulation
- ❸ AI should be avoided
- ❹ AI has no limitations

ANSWER: B

DETAILED EXPLANATION:

- AI is powerful but risky
- Understanding limitations crucial
- Responsible governance essential
- Balance innovation with risk management
- Ethics, regulation, and governance all needed

EXAM SUCCESS: Understand [classical AI](#), [neural networks](#), [ML types](#), and [AI risks](#)!